END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

B Tech CSAI/ECEAI, 2nd Semester

(AUGUST, 2022)

Subject Code: BAS 108	Subject: Probability and Statistics

Time:3 Hours | Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (5*4 =20)

(a) A and B alternately throw a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, show that his chance of winning is 30/61.

(b) Two independent random variables X and Y is said to follow binomial distribution with $n_1 = 8,\ p_1 = 1/3,$ and $n_2 = 5,\ p_2 = 1/3.$ Find the probability of $X + Y \geq 1.$

(c) Three balls are drawn at random without replacement from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn, find the joint probability distribution of (X ,Y).

(d) A company has the head office at Kolkata and a branch at Mumbai. Thepersonnel director wanted to know if the workers at the two places wouldlike the introduction of a new plan of work and a survey was conductedfor this purpose. Out of a sample of 500 workers at Kolkata, 62%favored the new plan, At Mumbai out of a sample of 400 workers, 41%were against the new plan. Is there any significant difference between thetwo groups in their attitude towards the new plan at 5% level of significance?

Given $z_{0.05} = -1.645,\ z_{0.025} = -1.96.$

UNIT-I

Q2 (6+4=10)

(a) Calculate the Median for thefollowing frequency distribution of marks.

Marks	0 - 8	8 -16	16-24	24 -32	32-40	40 -48	48-56
No. of Students	6	12	20	42	36	24	18

(b) A continuous random variable X that can assume any value between x=2 and x=5 has a density function given by $f(x) = k(1 + x)$. Find $P(X < 4).$

Q3 (3+3+4=10)

A random variable X has the following probability distribution.

X	0	1	2	3	4	5	6	7
P(x)	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find

(i) the value of k,

(ii) $P(1.5 < X < 4.5 \,/\, X > 2)$ and

(iii) the smallest value of λ for which $P(X \leq \lambda) > 1/2.$

UNIT-II

Q4 (10)

The joint probability density function of two random variables X and Y is given by:

$$f(x,y) = \begin{cases} 15x^2y\,; 0 < x < y < 1 \\ 0\ \ ;\ \text{elsewhere} \end{cases}$$

Find the correlation coefficient between $2X + 3$ and $-3Y + 5$

Q5 (5+5=10)

The joint probability density function of two random variables X and Y is given by:

$$f(x,y) = \begin{cases} cx + 1\,;\ \ x, y \geq 0;\ x + y < 1 \\ 0\ \ ;\ \text{elsewhere} \end{cases}$$

(i) Find the constant c.

(ii) Find the marginal density function of X.

(iii) Find the conditional distribution of Y given X=x.

UNIT-III

Q6 The marks of 5 randomly selected students in a mid term examination (x) and in the final examination (y) are as follows: **(10)**

x	70	50	50	80	70
y	80	70	60	40	30

Estimate the linear regression equation of Y on X. Hence, estimate the marks in the final examination of a student who has received 40 marks in the midterm examination.

Q7 (a) Derive the moment generating function of a Poisson Distribution with parameter λ. **(6+4=10)**

(b) The time it takes to make a question paper is normally distributed with mean 40 minutes andstandard deviation 15 minutes. Find the probability that one can finish making a questionpaper in less than 20 minutes.

UNIT-IV

Q8 The following table gives the number of good and bad parts produced by each of the three shifts in a factory: **(10)**

	Good Parts	Bad Parts
Day Shift	960	40
Evening Shift	940	50
Night Shift	950	45

Test whether or not the production of bad parts is independent of the shift on which they were produced at the 5% level of significance.
Given $\chi^2_{0.05} = 5.991$ (at 2df), $\chi^2_{0.05} = 9.488$ (at 4df).

Q9 Two independent samples of 8 and 7 items respectively had the following values of the variable (weight in ounces): **(10)**

Sample1	*9*	*11*	*13*	*11*	*15*	*9*	*12*	*14*
Sample 2	*10*	*12*	*10*	*14*	*9*	*8*	*10*	

Is the difference between the means of the sample significant? Given $t_{0.05} = 2.16$ (at 13 df), $t_{0.05} = 2.145$ (at 14df).

Cumulative Standard Normal Distribution

F(-0.87) = 0.19215, F(-1.34) = 0.09012, F(1.34) = 0.90988, F(0.87) = 0.80785